

C2_02 Gate B Protected Review - Astra RC4

Support SLA / reopen analysis. Open only after a saved independent Gate B attempt; then close the review and perform changed repair proof.

1. Objective controls

Control	Expected	Why
Tickets	100	ticket grain
Resolved tickets	60	ticket status control
Events	164	event grain
Resolved events	61	includes orphan event 9999
Team IDs	11 raw / 10 unique	team_id 3 conflicts
Reopened <=48h	3	tickets 51,68,85

2. Expected reasoning path

Start with two grains: tickets is one row/ticket; ticket_events is one row/event. A naive event join can duplicate ticket denominators.

Duplicate key: team_id 3 maps to Tier1 and Escalations and is used by tickets. Therefore team_type KPI output is not trustworthy until the mapping conflict is resolved or explicitly quarantined.

Event control: there are 60 tickets whose status is resolved but 61 resolved events because ticket_id 9999 is an orphan event. Event counts and ticket counts answer different questions.

Reopen contract: define reopened-within-48h from event timestamps after a valid resolved event. Tickets 51/team4, 68/team5 and 85/team6 qualify. At team_id grain, each team has 1 of 10 resolved tickets = 10%; that is a tie, not a unique worst team.

Weekly LAG: construct/check weekly series and denominator before comparing prior week. Missing weeks change what LAG means.

3. Critical failures

- Uses INNER JOIN to make orphans disappear without an explicit exception population.
- Uses DISTINCT as a generic fan-out repair instead of proving key/cardinality.
- Calculates a KPI at event grain while calling it ticket/order/invoice grain.
- Uses LAG before proving complete/deterministic time grain.
- Ranks a dirty dimension attribute before resolving a conflicting duplicate key.
- Reports an attractive total without independent row/distinct-key/money controls.
- Treats EXPLAIN or fast runtime as proof that business logic is correct.

4. Competency floors

Competency	Required proof
Grain/cardinality	Input/output grain; duplicate and orphan treatment; no unexplained fan-out.
SQL architecture	Readable CTE/subquery stages with controls at the correct grain.
Window/time	Partition/order/frame or LAG contract; deterministic period logic and gap awareness.
Validation	Rows, distinct keys, numerator/denominator and money independently reconcile.
Performance/handoff	Safe EXPLAIN evidence after correctness; query comments/assumptions and rerun path.

Changed repair proof: after review, close this PDF and solve a different grain/key/time case. Do not use the same Gate fixture as repair evidence. Gate C opens only when explicitly assigned.